WORKSHEET — UNIT I

Introduction to IoT

Course: IoT with Applications
Unit: I (3 Hrs / 180 min)
Topics: What is IoT?; IoT vs Related Concepts; Building Blocks of IoT; IoT Architecture; IoT Devices; The IoT Data Journey

🎯 Section A — Learning Objectives

1. Define IoT, list its characteristics, and trace its evolution from RFID tagging to AI-driven IoT.

2. Differentiate IoT from the Internet, M2M (Machine-to-Machine) and Cyber-Physical Systems.

3. Identify the components and enabling technologies that make up an IoT system.

4. Compare the three-layer and five-layer IoT architectures and explain the role of each layer, including edge and fog computing.

5. Classify IoT devices — sensors, actuators, microcontrollers, embedded systems, RFID/NFC/GPS and smart devices.

6. Describe the IoT data journey from data acquisition to real-time monitoring and dashboards.

🔑 Section B — Key Concepts

No.	Concept	Short Explanation
1	**Definition of IoT**	A network of physical objects with embedded sensors, software and connectivity that collect and exchange data with little human intervention.
2	**IoT vs M2M**	M2M is direct device-to-device communication, often on closed networks; IoT uses IP/internet connectivity, cloud platforms and analytics.
3	**Cyber-Physical System (CPS)**	Tight integration of computation with physical processes through feedback control; IoT often supplies the connectivity for CPS.
4	**Building Blocks of IoT**	Devices, connectivity, data processing and the user interface work together to form a complete IoT system.
5	**Three-Layer Architecture**	A simple model made of the Perception, Network and Application layers.
6	**Five-Layer Architecture**	An extended model: Perception, Transport/Network, Processing, Application and Business layers.
7	**Sensors vs Actuators**	Sensors convert physical quantities into electrical signals; actuators convert electrical signals into physical action.
8	**Edge & Fog Computing**	Data is processed near the source (edge) or in an intermediate layer (fog) to cut latency and bandwidth use.
9	**The IoT Data Journey**	Data flows from acquisition to transmission, storage, analytics and finally real-time monitoring on dashboards.

📘 Section C — Important Terminology

Keyword	Meaning
IoT	A network of interconnected physical objects that sense, communicate and act on data over the internet.
Sensor	A device that detects a physical quantity (e.g. temperature, light) and converts it into an electrical signal.
Actuator	A device that converts an electrical signal into physical action such as motion, heat or opening a valve.

Microcontroller	A compact chip with a processor, memory and input/output pins used to control a device.
Embedded System	A computer system built into a larger device to perform a dedicated function.
RFID	Radio-Frequency Identification: identifies and tracks objects using tags that respond to radio waves.
NFC	Near-Field Communication: very short-range wireless technology used for contactless payments and pairing.
GPS	Global Positioning System: satellite-based service that provides location and time information.
M2M	Machine-to-Machine: direct communication between devices without human involvement.
Cyber-Physical System	A system in which computing and networking tightly control physical processes.
Edge Computing	Processing data at or near the device where it is generated instead of in a distant cloud.
Fog Computing	An intermediate layer between edge devices and the cloud that provides local processing and storage.
Smart Device	A connected device with sensing, processing and communication abilities that can act intelligently.
Data Acquisition	The process of sensing, measuring and collecting data from the physical environment.
Dashboard	A visual interface of charts, gauges and alerts used to monitor IoT data in real time.

✅ Section D — Multiple-Choice Questions [Easy–Medium]

1. Which of the following best describes the Internet of Things (IoT)?

A. A network of only smartphones and laptops
B. A network of physical objects with embedded sensors, software and connectivity that exchange data
C. A private intranet limited to one building
D. A programming language for embedded devices

2. Which technology is regarded as an early forerunner of IoT for identifying and tracking objects?

A. Bluetooth Low Energy
B. Blockchain
C. RFID
D. Digital twins

3. What is the key difference between IoT and M2M?

A. IoT typically uses IP/internet connectivity and cloud platforms, while M2M often uses point-to-point closed networks
B. M2M uses cloud analytics while IoT does not
C. IoT cannot involve machines
D. M2M always requires human interaction

4. Which building block of an IoT system lets the user view data and send commands to devices?

A. Sensor
B. Gateway
C. Cloud storage
D. User interface

5. In the three-layer IoT architecture, which layer senses and collects data from the physical environment?

A. Application layer
B. Perception layer
C. Network layer
D. Business layer

6. Which layer of the five-layer architecture is mainly responsible for storing, analysing and processing large volumes of data?

A. Processing layer
B. Perception layer
C. Transport layer
D. Business layer

7. Which device converts an electrical signal into physical action, such as turning a motor or opening a valve?

A. Sensor
B. RFID tag
C. Actuator
D. GPS receiver

8. Processing data close to where it is generated, to reduce latency and bandwidth use, is called:

A. Cloud storage
B. Data mining
C. Blockchain
D. Edge computing

9. Which technology provides location and time information to IoT devices using satellites?

A. NFC
B. GPS
C. RFID
D. Zigbee

10. What is the correct order of the IoT data journey?

A. Storage → Acquisition → Analytics → Transmission
B. Transmission → Acquisition → Storage → Analytics
C. Acquisition → Transmission → Storage → Analytics
D. Analytics → Storage → Transmission → Acquisition

✏️ Section E — Fill in the Blanks [Easy]

1. The _____ layer of the IoT architecture contains the sensors and actuators that interact with the physical world.

2. _____ computing extends the cloud to an intermediate layer between edge devices and the cloud.

3. A _____ converts a physical quantity such as temperature or light into an electrical signal.

4. _____ is a very short-range wireless technology (a few centimetres) used in contactless payments.

5. A graphical interface that displays real-time IoT data using charts and gauges is called a _____.

☑️ Section F — True or False [Easy]

1. IoT devices always need a human to operate them manually. (True / False)

2. RFID is considered one of the earliest technologies that led to IoT. (True / False)

3. An actuator is used to measure physical quantities such as temperature. (True / False)

4. In the three-layer architecture, the Application layer delivers services to the end user. (True / False)

5. Edge computing sends all raw data to the cloud before any processing is done. (True / False)

🔗 Section G — Match the Following [Medium]

Column A	Column B (shuffled)
1. Sensor	a. Satellite-based location and time technology
2. Actuator	b. Chip with processor, memory and I/O that controls a device
3. Microcontroller	c. Visual interface that displays IoT data
4. RFID	d. Processing in an intermediate layer between edge and cloud
5. GPS	e. Converts a physical quantity into an electrical signal
6. Dashboard	f. Converts an electrical signal into physical action
7. Fog computing	g. Identifies objects using radio-frequency tags

📝 Section H — Short-Answer Questions [Medium]

1. Define IoT and list any two of its characteristics.

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2. What is the difference between IoT and M2M?

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3. List the four building blocks (components) of an IoT system.

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4. Differentiate between a sensor and an actuator, giving one example of each.

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5. What is the purpose of an IoT dashboard?

📄 Section I — Descriptive Questions [Medium–Advanced]

1. Explain the characteristics of IoT and describe its evolution from RFID tagging to today's AI-driven IoT.

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2. Compare IoT with the Internet, M2M and Cyber-Physical Systems.

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3. Describe the three-layer and five-layer IoT architectures with a neat diagram, and give a brief overview of edge and fog computing.

4. Explain the IoT data journey — from data acquisition to real-time monitoring — using a suitable example.

💻 Section J — Practical / Application-Based Questions [Advanced]

Practical 1

Problem: Design the sensing and control part of a Smart Room Temperature Controller.

Task:

1. List two sensors the system needs and state what each measures.

2. List two actuators the system needs and state what each controls.

3. Name the microcontroller-based unit that connects them and the layer of the architecture each part belongs to.

Expected Output: *A table of sensors and actuators (e.g. temperature sensor, occupancy sensor, fan relay, AC controller) with their roles and architecture layers.*

Practical 2

Problem: Draw the block diagram of a Smart Plant Watering System that keeps soil moisture in a safe range.

Task:

1. Identify the device, connectivity, data-processing and user-interface blocks of the system.

2. Map the blocks to the three-layer architecture.

3. Draw the block diagram showing the direction of data flow and the control action.

Expected Output: *A labelled block diagram: soil-moisture sensor → microcontroller → network → cloud/processing → app, with a pump (actuator) driven by the decision.*

Practical 3

Problem: Simulate a small IoT data journey in Python using temperature readings.

Task:

1. Store 8 temperature readings (data acquisition) in a list.

2. Calculate the average temperature (data analytics).

3. Print an ALERT message if the average exceeds 30°C, otherwise print “Normal” (real-time monitoring).

Expected Output: *The list of readings, the computed average, and either an ALERT or a Normal status message.*

🗺️ Section K — Concept-Mapping Activity

INTERNET OF THINGS (IoT)

Complete the map below: fill in the missing sub-topic name or example for each branch.

Branch 1	Branch 2	Branch 3	Branch 4
IoT Architecture	**IoT Devices**	**Data Journey**	**IoT vs Related Concepts**
Example: _____	Example: _____	Example: _____	Example: _____

(Answer key — example values: Perception, Network, Application layers; sensors, actuators, RFID; acquisition → transmission → storage → analytics; M2M, Cyber-Physical Systems)

🏢 Section L — Case Study / Real-World Activity

Scenario

A college wants to build a Smart Classroom system. Temperature, humidity and occupancy sensors will be placed in each room. The system should switch the lights and air-conditioner on or off automatically, and the facility team should be able to watch room conditions live on a dashboard and receive an alert if a room stays too hot.

Questions

1. Which sensors and actuators would you use, and what would each one do?

2. Map the system to the three-layer IoT architecture: what happens in each layer?

3. Trace the data journey of a temperature reading from acquisition to the dashboard.

4. Would you use edge computing to switch off the lights when a room becomes empty? Justify your answer.

Expected Learning Outcome

Students connect IoT building blocks, architecture layers, devices and the data journey to a realistic campus-automation problem.

📊 Section M — Self-Assessment

☐ **Yes** ☐ **Partly** ☐ **No** I can explain the major concepts of this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can define the important terminology from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can solve problems/exercises from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can apply the concepts of this unit to a new situation.

💭 Section N — Student Reflection

What did I learn in this unit?

What was difficult for me?

What do I need to practice more?

One important concept I learned:

🗝 Section O — Instructor Answer Key

(For instructor use — covers all objective and subjective items in this worksheet.)

MCQ Answers

- 1. B — IoT connects everyday physical objects that sense, communicate and share data over a network.
- 2. C — RFID tagging was the starting point of IoT; the term itself arose from RFID-based supply-chain tracking.
- 3. A — M2M is device-to-device, often on proprietary links; IoT is internet-based and integrates with cloud, analytics and apps.
- 4. D — The user interface (mobile app, web app or dashboard) is how people monitor and control the system.
- 5. B — The Perception layer holds sensors and actuators that interact directly with the physical world.
- 6. A — The Processing (middleware/cloud) layer stores, analyses and processes the data received from the network.
- 7. C — An actuator performs the physical action; a sensor does the opposite by measuring.
- 8. D — Edge computing moves processing to or near the device, avoiding a long round trip to the cloud.
- 9. B — GPS receivers calculate position and time from satellite signals.
- 10. C — Data is first acquired, then transmitted, stored and finally analysed.

Fill in the Blanks — Answers

- 1. Perception
- 2. Fog
- 3. sensor
- 4. NFC
- 5. dashboard

True/False — Answers

- 1. False
- 2. True
- 3. False
- 4. True
- 5. False

Match the Following — Answer Key

1 → e 2 → f 3 → b 4 → g 5 → a 6 → c 7 → d

Short Answer — Key Points

- 1. IoT is a network of physical objects with embedded sensors, software and connectivity that exchange data over the internet. Characteristics: connectivity, sensing, intelligence/analytics, scalability, automation (any two).
- 2. M2M is direct device-to-device communication, often over closed/cellular links; IoT is internet-based, uses cloud and analytics, and connects devices to applications and people.
- 3. Devices (sensors/actuators), connectivity (network), data processing (edge/cloud), and the user interface (app/dashboard).
- 4. A sensor measures a physical quantity (e.g. temperature sensor); an actuator performs a physical action (e.g. relay-driven fan or motor).
- 5. It presents live and historical IoT data as charts, gauges and alerts so users can monitor conditions and make decisions quickly.

Descriptive Questions — Key Points

- 1. Characteristics: connectivity, sensing, intelligence, automation, scalability. Evolution: RFID tagging → connected sensors/M2M → internet-connected devices and cloud → AI-driven IoT.
- 2. Internet connects people/computers; M2M is direct machine communication; CPS integrates computation with physical control; IoT connects things to the internet and cloud analytics. Present as a comparison table.

- 3. Three layers: Perception, Network, Application. Five layers: Perception, Transport, Processing, Application, Business. Describe each layer's role; edge = local processing at device, fog = intermediate layer, cloud = central.
- 4. Acquisition (sensors) → transmission (network/gateway) → storage (cloud/database) → analytics → real-time monitoring and dashboards; support with an example such as smart home temperature monitoring.

Practical Activities — Expected Outputs

- 1. Sensors: temperature, occupancy. Actuators: fan/AC relay, light switch. Controller: microcontroller board. Sensors/actuators belong to the Perception layer, communication to the Network layer, and the monitoring app to the Application layer.
- 2. Soil-moisture sensor → microcontroller → Wi-Fi/network → cloud processing → mobile app; pump (actuator) switches on when moisture falls below the set level.
- 3. Sample logic: readings = [28, 29, 31, 32, 30, 33, 29, 31]; avg = sum(readings)/len(readings); print("ALERT" if avg > 30 else "Normal"). Average here is 30.375, so the output is ALERT.

Concept Map — Sample Answers

- Architecture: Perception, Network, Application layers. Devices: sensors, actuators, RFID. Data Journey: acquisition, transmission, storage, analytics. Related Concepts: M2M, Cyber-Physical Systems.

Case Study — Key Points

- 1. Sensors: temperature, humidity, occupancy (PIR). Actuators: relay for lights, AC controller.
- 2. Perception: sensors and actuators; Network: Wi-Fi/gateway carrying data; Application: dashboard, rules and alerts.
- 3. Sensor reads temperature → microcontroller sends it over the network → stored in the cloud → analysed against a threshold → shown live on the dashboard.
- 4. Yes — switching lights needs a fast local response and works even if the internet is down, so edge processing reduces latency and bandwidth.